Sumit Jahagirdar

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I build agentic AI systems on enterprise data, and the harnesses that keep them reliable. My path through deep learning, software engineering, MLOps and AI solutions consulting keeps me grounded and flexible. I pick up new tech stacks and domains fast. I can tell early what AI can and can't do for a problem. I also scope where functional teams are critical to take prototypes live.

EXPERIENCE

Lead Data Scientist, Aays, Bengaluru Jun 2022 - Present

I architect solutions in presales, scope them with clients, and lead a pod through delivery.

- **Agent Factory: Multilingual Data Agent, DOCOMO Group.** Answers advanced analytics questions for DOCOMO telemarketing, in Japanese and English. Built for 400+ users across 166+ tables and 1,100+ columns. It runs on Databricks.
  - Agent harness I built from scratch. Tools run as an async DAG. The agent generates and executes code live.
  - Fine-tuned embeddings handle retrieval across both languages. Guardrails, lineage and evals are built in. *Stack:* Python, asyncio, Databricks, Spark, Delta Lake, Transformers.
- **Trade Fund Assistant, P&G ANZ and Japan.** Answers trade fund questions for sales teams. A LangGraph agent with text-to-SQL tools, calibrated templates, conversational memory and secure tool access. *Stack:* Python, LangGraph, LangChain, SQL.
- **Financial Document Parser, 73 Strings (private equity).** Automates financial data extraction for private equity analysts, reducing manual effort by ~90%. YOLO routes each page to the right OCR and table models. Inspectable rule engines clean numbers, units and currencies. Ensemble disagreement flags uncertain values for human review. *Stack:* Python, PyTorch, YOLO, LayoutLM, OpenCV.
- **Credit Agreement Parser.** Transforms 200+ page credit agreements into entities in a knowledge graph by identifying clauses, tracing cross-references between provisions, and normalizing clause structures for downstream analysis. *Stack:* Python, Neo4j.
- **Call Analyzer.** Scores outbound sales calls against their SOP. Whisper transcription and prompt-based rule engines, served efficiently on vLLM. *Stack:* Python, Whisper, vLLM.
- **Cricket Ball Tracking.** Speed and bounce point from phone video. Custom CNNs detect the ball. Signal processing cleans its path per delivery. *Stack:* Python, PyTorch, OpenCV, NumPy, SciPy.

Software Engineer, Edison AI, GE HealthCare, Bengaluru Aug 2021 - May 2022

- Built microservices and a Python SDK on AWS Lambda and Kubernetes. *Stack:* Python, AWS Lambda, Docker, Kubernetes.

Machine Learning Research Intern, NVIDIA Apr 2019 - Apr 2020

- Presented few-shot medical imaging work at GTC 2020. *Stack:* Python, PyTorch.

OPEN SOURCE

- **CodePulse** (github.com/sumitrj/CodePulse). Shows coding agents which files matter. A live tree-sitter call graph served over MCP, benchmarked on SWE-bench. *Stack:* Python, tree-sitter, MCP SDK.

SKILLS

AI core: deep learning (PyTorch, Transformers), NLP, computer vision (YOLO, OpenCV), speech and signal processing (Whisper/ASR), fine-tuning (LLM/CNN, embeddings), OCR, knowledge graphs

Agents & LLMs: AI agents, agent harnesses, RAG, MCP, evals, guardrails, LangGraph, OpenAI SDK, vLLM, text-to-SQL

Engineering: Python, FastAPI, Databricks/Spark, MLflow, Delta Lake, AWS (Lambda), Kubernetes, tree-sitter

EDUCATION

B.Tech, Electronics and Communication Engineering (IoT), VIT Vellore, 2017-2021 (CGPA 8.03). Technical Head, IET VIT; Microsoft Student Ambassador (Top 30, APAC).